Problem 1: A two-cycle quantum conveyor: bands, projectors and flavors

A.1 **1.5 pt**

B.1(a) **2.7 pt**

B.1(b)

B.1(c)

B.2 **1.4 pt**

C.1 **2.3 pt**

D.1(a) **2.1 pt**

D.1(b)

D.1(c)

Problem 2: Correlations and the physical current

A.1 **1.3 pt**

B.1 **1.8 pt**

B.2 **0.8 pt**

C.1(a) **1.8 pt**

C.1(b)

C.1(c)

C.1(d)

C.2(a) **1.3 pt**

C.2(b)

C.2(c)

D.1 **0.9 pt**

D.2 **2.1 pt**

Problem 3: How long can the conveyor operate?

A.1 **1.1 pt**

A.2 **2.2 pt**

A.3(a) **2.0 pt**

A.3(b)

A.3(c)

B.1 **1.6 pt**

C.1(a) **1.7 pt**

C.1(b)

C.1(c)

C.2 **1.4 pt**